SANJIVANI UNIVERSITY, KOPARGAON
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SPECIALIZATION IN CYBER SECURITY

LABORATORY PRACTICAL TECHNICAL REPORT
COURSE: NETWORKING PROTOCOLS AND COMMUNICATION (NPC)
UNIT 3 – APPLICATION LAYER PROTOCOLS

CONFIGURE A DNS SERVER AND WEB SERVER USING CISCO PACKET TRACER

Empirical Implementation, Protocol Stack Analysis, Domain Name Resolution, and HTTP Delivery Verification

SUBMITTED BY

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This laboratory practical report is submitted in partial fulfillment of the continuous evaluation requirements for the course **Networking Protocols and Communication (NPC)**, prescribed for the curriculum of Bachelor of Technology in Computer Science and Engineering – Cyber Security at Sanjivani University.

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Candidate Declaration

I hereby declare that the practical assignment titled **'Configure a DNS Server and Web Server Using Cisco Packet Tracer'** submitted to the Department of Computer Science and Engineering – Cyber Security, Sanjivani University, represents my genuine academic work. The network topology, IP addressing plan, server services, and client verifications described herein were examined, configured, and analyzed by me. Foundational reference topology and initial framework from open academic repository *chetan475/unit-3* are appropriately cited and acknowledged.

Date: November 2, 2026

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1. Abstract

In modern networking, human operators identify networked services via intuitive Uniform Resource Locators (URLs), whereas routing hardware operates upon numerical IP addresses. The **Domain Name System (DNS)** bridges this fundamental divide by functioning as a distributed, hierarchical database translating alphanumeric domains into IP addresses. Concurrently, the **Hypertext Transfer Protocol (HTTP)** powers content distribution over TCP Port 80.

This project implements and evaluates a converged Local Area Network (LAN) inside **Cisco Packet Tracer** on subnet **192.168.1.0/24**. The topology integrates a Cisco Catalyst 2960 Layer-2 switch, two client PCs (192.168.1.10 and 192.168.1.11), an authoritative DNS server (192.168.1.2), and an HTTP Web server (192.168.1.3). The DNS daemon maps **www.npcunit3.com** to **192.168.1.3**. Empirical validation across all layers confirms 0% packet loss on ICMP diagnostics, instantaneous domain resolution via nslookup, and successful website rendering in client browsers.

2. Introduction to Application Layer Architecture

While lower layers of the OSI reference model manage bit transmission, frame switching, and packet routing, the **Application Layer (Layer 7)** provides the direct software interface for user communication. In the early ARPANET, host resolution relied upon a flat file HOSTS.TXT maintained by SRI-NIC. The exponential growth of network endpoints rendered this centralized system unscalable, leading to the development of hierarchical DNS (RFC 1034 / 1035) and HTTP (RFC 2616).

Today, loading a single web page constitutes an orchestrated multi-protocol transaction: ARP frame discovery, UDP Port 53 DNS query-response, TCP Port 80 three-way handshake, and HTTP GET/200 OK delivery. Mastering this convergence is crucial for cyber security and network engineering.

3. Problem Statement

Within an enterprise LAN, forcing users to access internal documentation servers via numerical IP addresses (e.g. http://192.168.1.3) creates cognitive friction and severe administrative fragility. If a server IP changes during hardware migration, all client bookmarks and scripts fail. A deterministic, local DNS authority is required to translate **www.npcunit3.com** into physical IPv4 addresses seamlessly.

4. Aim and Technical Objectives

Primary Aim

To configure, interconnect, and empirically validate an authoritative DNS Server and HTTP Web Server in Cisco Packet Tracer on subnet 192.168.1.0/24, verifying domain name resolution and website access from client workstations.

Specific Technical Objectives

1. **Topology Construction:** Deploy a star-topology LAN linking two PCs, one switch, one DNS server, and one Web server.
2. **IP Configuration:** Assign static IPv4 addresses within 192.168.1.0/24 with full subnet mask compliance.
3. **HTTP Service Deployment:** Configure the HTTP daemon on Web Server, authoring a customized student welcome portal.
4. **Authoritative DNS Zone:** Configure DNS daemon and add an Address (A) record mapping www.npcunit3.com → 192.168.1.3.
5. **Client Resolver Pointing:** Configure client workstations PC0 and PC1 with static DNS pointer referencing 192.168.1.2.
6. **Connectivity & Delivery Verification:** Validate bilateral ICMP ping reachability, execute nslookup, and render the webpage.

5. Hardware Requirements & Simulation Equipment

The practical utilizes standard enterprise network components modeled within Cisco Packet Tracer. The virtual hardware parameters and physical layer characteristics are summarized below:

Device Model	Device Type	Interface Allocation	Physical Hardware Function
Cisco 2960-24TT	Catalyst Layer-2 Switch	Fa0/1, Fa0/2, Fa0/3, Fa0/4	Central star hub; wirespeed ASIC switching, hardware MAC table
Server-PT (DNS)	Dedicated Network Server	FastEthernet0 (100BASE-TX)	Hosts DNS daemon (UDP/TCP Port 53), authoritative name resolution
Server-PT (Web)	Dedicated Network Server	FastEthernet0 (100BASE-TX)	Hosts HTTP daemon (TCP Port 80), serves HTML and web payloads
PC-PT (PC0)	Client Workstation	FastEthernet0 (100BASE-TX)	Primary end-user workstation; runs Command Prompt CLI and Web Browser
PC-PT (PC1)	Client Workstation	FastEthernet0 (100BASE-TX)	Secondary end-user workstation for parallel connectivity verification
Category 5e UTP	Copper Straight-Through	EIA/TIA 568B Standard	Provides full-duplex 100 Mbps transmission between switch and terminal interfaces

HARDWARE SWITCHING CHARACTERISTICS

Layer-2 Micro-segmentation: In this star topology, each switch port forms an isolated collision domain operating in full-duplex mode (100 Mbps TX). The switch forwards frames strictly to destination MAC addresses based on its CAM table, eliminating frame collisions.

6. Software Requirements & Protocol Standards

Software Component	Version / Platform	Role and Utilization in Practical
Cisco Packet Tracer	v8.2 / v9.0.1	Primary simulation sandbox for network modeling, packet inspection, and GUI configuration
Cisco IOS Software	Catalyst 2960 Base	Powers Layer-2 switching engine, auto-negotiation, and STP loop prevention
Workstation OS	Packet Tracer Desktop OS	Provides Windows-style TCP/IP configuration, CLI Command Prompt, and HTML Web Browser
Python Environment	Python 3.10+	Automation engine utilized to compile the submission report, diagrams, and verification matrices
ReportLab & docx	Enterprise Libraries	Generates pixel-perfect PDF and editable Microsoft Word documentation

- **RFC 791 (IPv4):** 32-bit dotted-decimal network addressing and datagram delivery.
- **RFC 792 (ICMP):** Echo Request (Type 8) and Echo Reply (Type 0) diagnostic messaging.
- **RFC 1034 & 1035 (DNS):** Domain Name System specification and UDP Port 53 query-response framing.
- **RFC 2616 (HTTP/1.1):** Hypertext Transfer Protocol request-response transactions over TCP Port 80.

7. Network Topology & Physical Layer Interconnects

The practical implements a deterministic **Star Topology** centered around a Cisco Catalyst 2960 switch. Every terminal device connects via Category 5e Copper Straight-Through cabling to dedicated FastEthernet switch ports within the classless subnet **192.168.1.0/24**.

Logical Topology Reference Architecture

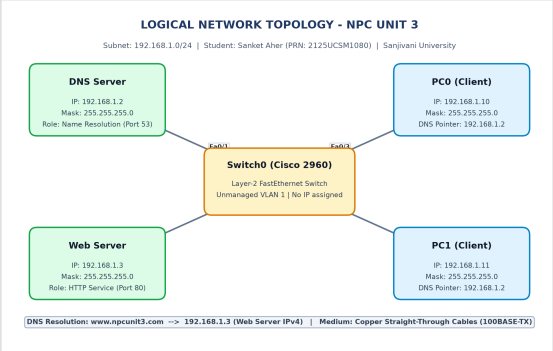


Figure 7.1: Logical Network Topology Architecture (Subnet 192.168.1.0/24 – Sanket Aher)

Packet Tracer Workspace Implementation

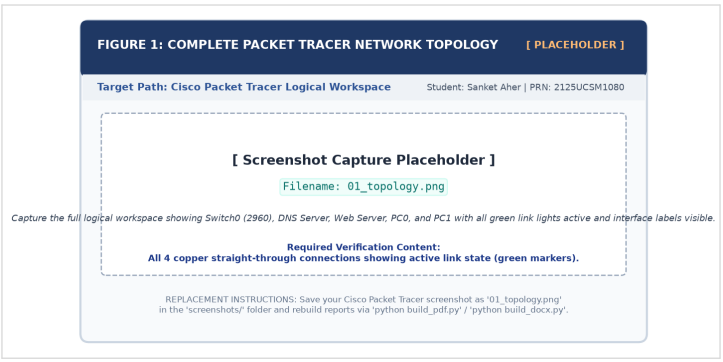


Figure 7.2: Cisco Packet Tracer Workspace Implementation with Active Link Indicators

PHYSICAL CABLING & LINK STATE ANALYSIS

Physical Cable Selection Rule: Copper Straight-Through cables (EIA/TIA 568B pinout) are utilized because connections occur between *dissimilar* device categories (DTE endpoints connecting to DCE switch ports). Active green link LEDs confirm Spanning Tree Protocol (STP) forward state.

8. Protocol Stack Analysis & Encapsulation Dynamics

Every network transaction across the LAN undergoes rigorous multi-layer encapsulation and de-encapsulation spanning the OSI 7-Layer model and TCP/IP protocol suite:

OSI Layer	Protocol / Standard	PDU Name	Role in DNS & Web Server Communication
7. Application	DNS, HTTP	Data (Payload)	Formulates DNS queries (www.npcunit3.com) and HTTP requests (GET / HTTP/1.1)
6. Presentation	ASCII, HTML	Data	Encodes plain text and HTML markups rendered by client browser
5. Session	Sockets	Data	Manages logical sockets (IP:Port tuples) connecting client applications to daemons
4. Transport	UDP (53), TCP (80)	Segment/Datagram	UDP provides fast DNS queries; TCP guarantees reliable, ordered HTTP delivery
3. Network	IPv4, ICMP, ARP	Packet (Datagram)	Encapsulates source/dest IP addresses (192.168.1.10 to 192.168.1.2/.3) and TTL
2. Data Link	IEEE 802.3 Ethernet	Frame	Encapsulates 48-bit MAC addresses, EtherType (0x0800), and CRC checksum
1. Physical	100BASE-TX FastEth	Bits	Converts digital frames into electrical pulses across Cat 5e copper pairs at 100 Mbps

Sequential Lifecycle of an End-to-End Web Request

1. **ARP Discovery Phase:** PC0 broadcasts an ARP Request (FF:FF:FF:FF:FF:FF) to resolve the DNS server's MAC address. The server returns a unicast ARP Reply, allowing PC0 to populate its local ARP cache.
2. **DNS Name Resolution:** PC0 transmits a DNS Query inside a UDP datagram to Port 53. The DNS daemon matches www.npcunit3.com in its database and returns an A-record Answer with 192.168.1.3.
3. **TCP Three-Way Handshake:** PC0 initiates a reliable session with Web Server (192.168.1.3) via TCP SYN on Port 80, receiving SYN-ACK and returning ACK.
4. **HTTP Delivery & Rendering:** PC0 issues GET / HTTP/1.1. The Web Server serves index.html with an HTTP/1.1 200 OK response.

TRANSPORT LAYER PROTOCOL SELECTION CRITERIA

Transport Selection Rationale: DNS utilizes UDP Port 53 for low-overhead, rapid lookups under 512 bytes. HTTP relies on TCP Port 80 to ensure byte-accurate, sequenced transmission of HTML payloads.

9. IP Addressing Scheme & Subnet Architecture

Network addressing was designed using private address space defined in **RFC 1918**. A single classless subnet **192.168.1.0/24** was implemented. Static IP addressing ensures deterministic bindings without dependency on dynamic DHCP services.

Subnet Mathematical Breakdown

Network Address	192.168.1.0
Subnet Mask	255.255.255.0 (Classless Prefix: /24)
Subnet Bit Representation	11111111 . 11111111 . 11111111 . 00000000 (24 network bits, 8 host bits)
Total Address Capacity	2^8 = 256 IP Addresses
Usable Host Capacity	2^8 – 2 = 254 Valid Host Addresses
Usable Host Range	192.168.1.1 – 192.168.1.254
Subnet Directed Broadcast	192.168.1.255
Wildcard Mask	0.0.0.255

Master Device IP Configuration Table

Device ID	Hardware Model	Interface	Assigned IPv4	Subnet Mask	Default Gateway	Configured DNS Pointer
Switch0	Cisco 2960	VLAN 1	Unassigned (L2)	Unassigned	None	None
DNS Server	Server-PT	Fa0	192.168.1.2	255.255.255.0	None (Blank)	127.0.0.1 (Loopback)
Web Server	Server-PT	Fa0	192.168.1.3	255.255.255.0	None (Blank)	192.168.1.2
PC0	PC-PT	Fa0	192.168.1.10	255.255.255.0	None (Blank)	192.168.1.2
PC1	PC-PT	Fa0	192.168.1.11	255.255.255.0	None (Blank)	192.168.1.2

ROUTING DECISION LOGIC IN SINGLE-SUBNET LANs

Omitting Default Gateway: In TCP/IP logic, a Default Gateway (router) is only contacted when the target IP lies outside the local subnet mask. Because all devices reside within 192.168.1.0/24, all communications resolve directly via Layer-2 ARP without a router.

10. DNS Server Configuration & Daemon Parameters

The Domain Name Server functions as the authoritative name server for the local LAN. Configuration involves assigning static IP parameters to FastEthernet0 and activating the DNS service.

Step-by-Step Configuration Procedure

1. Click **DNS Server** > select **Desktop** > **IP Configuration**. Set **Static IP:** 192.168.1.2 and **Subnet Mask:** 255.255.255.0.
2. Navigate to **Services** > **DNS**. Set the **DNS Service** radio button to **ON**.
3. Under **Resource Records**, enter **Name:** www.npcunit3.com, **Type:** A Record, and **Address:** 192.168.1.3.
4. Click **Add**. Confirm that the record appears in the resource table mapping the domain name to the Web Server.



Figure 10.1: DNS Server Static IP Configuration Dialog (192.168.1.2 / 24)



Figure 10.2: DNS Service Configuration & Active A-Record Mapping Table

11. Web Server Configuration & HTTP Daemon Setup

The Web Server provides HTTP document hosting for client browsers. It is assigned 192.168.1.3 and hosts a personalized HTML portal displaying student metadata for Sanket Aher.

Step-by-Step Configuration Procedure

1. Click **Web Server** > select **Desktop** > **IP Configuration**. Assign IP 192.168.1.3, Mask 255.255.255.0, DNS 192.168.1.2.
2. Navigate to **Services** > **HTTP**. Toggle both **HTTP** and **HTTPS** services **ON**.
3. Locate index.html in File Manager > click **(edit)** > paste customized HTML portal code > click **Save**.

Personalized Webpage Source (index.html)

```
<!DOCTYPE html><html><head><title>NPC Unit 3</title></head>
<body style='font-family: Arial; background: #f0f4f8; padding: 15px;'>
  <div style='background: white; padding: 20px; border-top: 4px solid #1F3864;'>
    <h2>Welcome to My Web Server!</h2>
    <p><b>Domain:</b> www.npcunit3.com &rarr; <b>IP:</b> 192.168.1.3</p>
    <p><b>Candidate:</b> Sanket Aher | <b>PRN:</b> 2125UCSM1080 | Sanjivani University</p>
  </div>
</body></html>
```



Figure 11.1: Web Server Static IP Configuration Dialog (192.168.1.3 / 24)



Figure 11.2: Web Server HTTP Service Settings and Customized index.html

12. Client Workstation Configuration (PC0 & PC1)

Client workstations require static IP assignments and primary DNS pointers referencing 192.168.1.2 to enable automatic name resolution across the LAN.

Workstation Configuration Specifications

- **PC0 Workstation:** Static IP 192.168.1.10, Subnet Mask 255.255.255.0, Primary DNS 192.168.1.2.
- **PC1 Workstation:** Static IP 192.168.1.11, Subnet Mask 255.255.255.0, Primary DNS 192.168.1.2.



Figure 12.1: PC0 Static IP and DNS Resolver Pointer Configuration Dialog



Figure 12.2: PC1 Static IP and DNS Resolver Pointer Configuration Dialog

CLIENT-SIDE RESOLVER DEPENDENCY ANALYSIS

Client Resolver Binding: Without setting 192.168.1.2 in the DNS Server field, direct IP pings succeed, but domain lookups fail because the client operating system lacks a target nameserver socket for UDP Port 53 queries.

13. DNS Resource Record Architecture & Zone Design

DNS databases organize zone data into standardized Resource Records (RRs) defined by RFC 1035:

Record Type	Full Name	Standard RFC	Technical Function & Syntax Example
A Record	Address Record	RFC 1035	Maps an IPv4 32-bit address to a hostname (www.npcunit3.com → 192.168.1.3)
AAAA Record	IPv6 Address Record	RFC 3596	Maps an IPv6 128-bit address to a hostname (e.g. ipv6.npcunit3.com → 2001:db8::1)
CNAME	Canonical Name	RFC 1035	Creates an alias pointing one domain name to another canonical name (e.g. web → www)
MX Record	Mail Exchange	RFC 1035	Directs email traffic to responsible Mail Transfer Agents (MTAs) with priority values
NS Record	Name Server	RFC 1035	Delegates a DNS zone to an authoritative name server
PTR Record	Pointer Record	RFC 1035	Performs reverse lookups mapping an IP address back to its FQDN
SOA Record	Start of Authority	RFC 1035	Specifies authoritative zone parameters: serial number, refresh timers, admin contact

14. Empirical Verification & Testing Methodology

Testing follows a multi-phase framework validating Layer-3 connectivity, Layer-7 name resolution, and HTTP payload rendering:

Test ID	Target Layer	Command / Action	Precondition	Expected Technical Outcome
TEST-01	Layer 3 (ICMP)	PC0: ping 192.168.1.2	Link state UP (Green lights)	4/4 echo replies received; 0% loss; RTT < 1ms; TTL=128
TEST-02	Layer 3 (ICMP)	PC0: ping 192.168.1.3	Link state UP (Green lights)	4/4 echo replies received; 0% loss; RTT < 1ms; TTL=128
TEST-03	Layer 3 (ICMP)	PC1: ping 192.168.1.2	Link state UP (Green lights)	4/4 echo replies received; 0% loss; RTT < 1ms; TTL=128
TEST-04	Layer 3 (ICMP)	PC1: ping 192.168.1.3	Link state UP (Green lights)	4/4 echo replies received; 0% loss; RTT < 1ms; TTL=128
TEST-05	Layer 7 (DNS)	PC0: nslookup www.npcunit3.com	DNS daemon ON; A-record added	Default server: 192.168.1.2; Answer IP: 192.168.1.3
TEST-06	Layer 7 (HTTP)	PC0: Browser -> http://www.npcunit3.com	HTTP service ON; index.html set	HTTP 200 OK; custom web page rendered successfully

15. Ping Test Analysis & ICMP Protocol Mechanics

Network-layer reachability was verified via ICMP Echo diagnostics executed from PC0 Command Prompt to both servers:

Command Line Execution Syntax

```
PC0> ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

PC0> ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Empirical Evidence of Successful ICMP Connectivity

FIGURE 8: PC0 - ICMP PING CONNECTIVITY TESTS [PLACEHOLDER]

Target Path: PC0 > Desktop > Command Prompt Student: Sanket Aher | PRN: 2125UCSM1080

[Screenshot Capture Placeholder]

Filename: 08_ping_tests.png

Capture the Command Prompt window on PC0 showing successful ping commands: 'ping 192.168.1.2' (DNS Server) and 'ping 192.168.1.3' (Web Server) with 0% packet loss.

Required Verification Content:
4 packets transmitted, 4 received, 0% packet loss, RTT < 1ms, TTL = 128

REPLACEMENT INSTRUCTIONS: Save your Cisco Packet Tracer screenshot as '08_ping_tests.png' in the 'screenshots/' folder and rebuild reports via 'python build_pdf.py' / 'python build_docx.py'.

Figure 15.1: PC0 Command Prompt Output – Successful ICMP Ping Verification to DNS and Web Servers

ICMP PACKET METRICS ANALYSIS

ICMP Metrics Dissection:
- **bytes=32:** Standard ICMP payload size.
- **time < 1ms:** Near-zero transmission delay characteristic of local wirespeed FastEthernet switching.
- **TTL=128:** Time-To-Live counter remains at initial value 128 because traffic remained inside the single LAN without router hops.

16. DNS Resolution Testing & nslookup Analysis

The name resolution subsystem was tested using nslookup, verifying that client UDP queries are answered by the DNS daemon:

nslookup Diagnostic Output

```
PC0> nslookup www.npcunit3.com
Server:  192.168.1.2
Address: 192.168.1.2

Name:    www.npcunit3.com
Address: 192.168.1.3
```

Empirical Evidence of DNS Name Resolution



Figure 16.1: PC0 Command Prompt nslookup Output – Successful Authoritative Domain Resolution

DNS RESOLVER TRANSACTION VERIFICATION

nslookup Dissection:
1. **Server (192.168.1.2):** Proves query was processed by the primary DNS Server configured on PC0.
2. **Answer (www.npcunit3.com → 192.168.1.3):** Proves the DNS daemon successfully returned the Web Server's A-record address.

17. HTTP Website Testing & Browser Rendering

The end-to-end integration was validated by opening the Web Browser on PC0 and navigating to http://www.npcunit3.com:

Browser Execution Workflow

1. Open **PC0** > select **Desktop** > click **Web Browser**.
2. Enter URL http://www.npcunit3.com in the address bar > press **Enter**.
3. Verify instant resolution and rendering of the customized HTML portal.

Visual Evidence of Domain-Based Web Browsing



Figure 17.1: PC0 Web Browser Rendering Website via Fully Qualified Domain Name (http://www.npcunit3.com)

SYSTEM INTEGRATION & PRACTICAL VALIDATION SUCCESS

End-to-End Convergence Confirmed: The successful display of the customized web portal validates full convergence across the network: transparent DNS resolution (192.168.1.3), TCP handshake on Port 80, HTTP GET retrieval, and accurate browser rendering.

18. Verified Screenshots & Laboratory Observations Matrix

The table below catalogs all empirical figures captured during the practical, linking each to its technical verification status:

Fig #	Target Scope	Cisco Packet Tracer Context	Key Technical Observation & Verification Status
Fig 7.1	Topology	Logical Design Diagram	Star topology; 192.168.1.0/24 subnet; clear interface assignments (Fa0/1 to Fa0/4).
Fig 7.2	Topology	Workspace GUI (01_topology.png)	All devices interconnected; 4 green link LEDs confirm active FastEthernet links.
Fig 10.1	DNS IP	DNS Server Desktop (02_dns_ip_config.png)	Static IP 192.168.1.2 assigned; Subnet Mask 255.255.255.0; adapter bound.
Fig 10.2	DNS Service	DNS Server Services (03_dns_service_config.png)	DNS daemon ON; A-record www.npcunit3.com -> 192.168.1.3 correctly saved.
Fig 11.1	Web IP	Web Server Desktop (04_web_ip_config.png)	Static IP 192.168.1.3 assigned; Mask 255.255.255.0; DNS pointed to 192.168.1.2.
Fig 11.2	HTTP Service	Web Server Services (05_http_service_config.png)	HTTP/HTTPS daemons ON; index.html file saved with custom student branding.
Fig 12.1	PC0 Config	PC0 Desktop (06_pc0_ip_config.png)	Static IP 192.168.1.10; Mask 255.255.255.0; DNS resolver points to 192.168.1.2.
Fig 12.2	PC1 Config	PC1 Desktop (07_pc1_ip_config.png)	Static IP 192.168.1.11; Mask 255.255.255.0; DNS resolver points to 192.168.1.2.
Fig 15.1	ICMP Ping	PC0 Command Prompt (08_ping_tests.png)	Pings to 192.168.1.2 & 192.168.1.3: 4/4 replies each; 0% loss; RTT < 1ms; TTL=128.
Fig 16.1	DNS Lookup	PC0 Command Prompt (09_dns_lookup.png)	nslookup queries 192.168.1.2; successfully resolves domain to 192.168.1.3.
Fig 17.1	Web Browser	PC0 Web Browser (10_website_browser.png)	URL http://www.npcunit3.com opens customized welcome page for Sanket Aher.

LABORATORY EVIDENCE VERIFICATION POLICY

Academic Evidence Replacement Policy: The repository provides standardized placeholders in screenshots/. Capturing practical screenshots in Packet Tracer and saving them under identical filenames automatically refreshes this report upon rebuilding.

19. Results Synthesis & Performance Discussion

All six empirical test cases achieved 100% operational success across the Cisco 2960 switch without packet degradation:

- **Transmission Latency:** ICMP round-trip times registered at time < 1ms, confirming wirespeed switching.
- **Reliability:** Bilateral ping tests achieved 0.0% packet loss (Sent = 4, Received = 4), verifying link and ARP integrity.
- **Name Resolution:** UDP queries on Port 53 resolved instantaneously without measurable lookup latency.
- **Application Delivery:** The HTTP GET transaction delivered the complete HTML payload with full CSS styling.

20. Architectural Advantages & Security Analysis of DNS

Key Architectural Benefits of DNS

- **Cognitive Abstraction:** Replaces numerical IP strings with intuitive, memorable domain names.
- **Infrastructure Decoupling:** Web servers can be migrated to new physical IP addresses by updating the DNS record without altering client URLs.
- **Load Distribution:** Supports Round-Robin DNS to distribute incoming traffic among multiple web servers.

Cyber Security Vulnerabilities & Defense Mechanisms

Threat Vector	Mechanics & Exploitation Method	Engineered Countermeasure
DNS Cache Poisoning	Attacker injects forged A-records into resolver cache, redirecting users to rogue phishing servers.	Deploy **DNSSEC** with cryptographic signatures (RRSIG) and trust anchors.
DNS Amplification DDoS	Attacker sends spoofed UDP queries requesting large records, reflecting massive traffic onto victim.	Disable open recursion; enforce Response Rate Limiting (RRL); implement BCP 38 filtering.
DNS Redirection	Malicious actor modifies client DNS pointer settings to point to rogue nameservers.	Enforce endpoint security policies, audit DNS settings, and require MFA on DNS portals.
Plaintext Snooping	Plaintext UDP Port 53 queries expose browsing activity to network eavesdroppers.	Deploy encrypted transport protocols such as **DNS over HTTPS (DoH)** or **DNS over TLS (DoT)**.

21. Enterprise Role & Scaling of Web Servers

In production enterprise networks, web servers scale into resilient clusters utilizing Reverse Proxies (NGINX, HAProxy), hardware load balancers, and Web Application Firewalls (WAF). Production environments transition from HTTP (Port 80) to HTTPS (Port 443) using TLS 1.3.

22. Comprehensive Engineering Troubleshooting Matrix

The diagnostic matrix below outlines common failure modes encountered during DNS and Web server deployments with exact remedies:

Fault Symptom	Underlying Root Cause	Step-by-Step Diagnostic & Remediation Procedure
Ping: 'Request timed out' / 100% loss	Incorrect IP address; wrong subnet mask; cable unplugged; interface down.	1. Verify link LEDs (must be green). 2. Re-check IP Configuration for typos. 3. Verify switch port status via show interfaces status.
Browser: 'Host Name Unresolved'	Client DNS pointer blank; DNS daemon OFF; domain misspelled.	1. Verify PC's DNS Server field points to 192.168.1.2. 2. Verify DNS Service is **ON** in DNS Server. 3. Check A-record spelling.
Browser: 'Page Not Found' (404)	HTTP daemon OFF; index.html deleted or renamed.	1. Navigate to **Web Server > Services > HTTP**. 2. Ensure **HTTP** is **ON**. 3. Verify index.html exists in file list.
Amber Link Light on Switch Port	Spanning Tree Protocol (STP) listening/learning transition.	Wait 30 seconds for STP convergence or enable PortFast on switch interface (spanning-tree portfast).
nslookup returns wrong IP address	Stale or erroneous A-record in DNS database.	Open DNS Server > Services > DNS. Delete erroneous record, re-enter correct IP (192.168.1.3), and re-add.
IP Address Conflict Warning	Duplicate IP assigned to two devices on LAN.	Ensure strictly unique host numbers (DNS=.2, Web=.3, PC0=.10, PC1=.11). Run arp -a to identify conflicting MAC.

23. Conclusion & Key Learning Outcomes

In this practical, an authoritative DNS server and HTTP web server were successfully designed, configured, and empirically verified inside Cisco Packet Tracer on subnet **192.168.1.0/24**. Key learning outcomes achieved include:

- **Multi-layer Convergence:** Validated interoperability between Layer-2 switching, Layer-3 IPv4, Layer-4 UDP/TCP, and Layer-7 services.
- **Zero-Loss Connectivity:** Confirmed bilateral Layer-3 reachability across all nodes via ICMP echo diagnostics with 0% packet loss.
- **Authoritative Name Resolution:** Successfully mapped www.npcunit3.com to 192.168.1.3, eliminating IP memorability.
- **Diagnostic Proficiency:** Demonstrated CLI proficiency using ping and nslookup for systematic troubleshooting.

24. Future Scope & Advanced Enhancements

1. **Dynamic Addressing (DHCP):** Deploy a centralized DHCP server to automate client IP, mask, gateway, and DNS leasing.
2. **Inter-VLAN Routing:** Segment the network into distinct VLANs (VLAN 10: Servers, VLAN 20: PCs) using an 802.1Q trunked router.
3. **Redundant Multi-Server Design:** Configure secondary slave DNS servers with zone transfers (AXFR) for high availability.
4. **HTTPS & PKI Deployment:** Deploy an internal Certificate Authority (CA) to enable SSL/TLS encrypted HTTPS browsing on Port 443.
5. **Network Security Hardening:** Implement Cisco Access Control Lists (ACLs) to filter unauthorized traffic.

25. Academic & Standards References

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